



Toward ambient temperature operation with all-solid-state lithium metal batteries with a sp^3 boron-based solid single ion conducting polymer electrolyte



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HIGHLIGHTS

- Charge delocalized sp^3 boron based single ion conducting polymer electrolyte was prepared.
- PEO/SIPE based all-solid-state lithium ion battery displays battery operation at 50 °C.
- The PEO/SIPE blended membrane presumably presented two Li^+ pathways.

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ABSTRACT

The ionic conductivity decay problem of poly(ethylene oxide) (PEO)-based solid polymer electrolytes (SPEs) when increase the lithium salt of the SPEs up to high concentration is here functionally overcome by the incorporation of a charge delocalized sp^3 boron based single ion conducting polymer electrolyte (SIPE) with poly(ethylene oxide) to fabricate solid-state sp^3 boron based SIPE membranes (S-BSMs). By characterizations, particularly differential scanning calorimeter (DSC) and ionic conductivity studies, the fabricated S-BSMs showed decreased melting points and increased ionic conductivity as steadily increase the content of sp^3 boron based SIPE, which significantly improved the low temperature performance of the all-solid-state lithium batteries. The fabricated Li | S-BSMs | LiFePO₄ cells exhibit highly electrochemical stability and excellent cycling at temperature below melting point of PEO, which has never been reported so far for SIPEs based all-solid-state lithium batteries.

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1. Introduction

All-solid-state lithium-ion batteries capable of operating at ambient temperatures have long been perceived as a holy grail in modern battery research and represent a significant challenge for materials and device development [1–4]. Successful fabrication of all-solid-state battery systems would provide an ultimate solution to the long standing technical challenges on device safety and dendrite formation, which have been intrinsically associated with Li-ion batteries with liquid and/or gel electrolytes for many years

[5]. An advance in development of all-solid-state battery cells is inherently dependent on solid polymer electrolytes (SPEs), such as poly(ethylene oxide) (PEO)/lithium salts, since high ionic conductivity and low interfacial resistance are required for the battery cells to operate [6]. PEO/lithium salt systems have been most extensively studied because PEO is capable of providing necessary flexibility to facilitate lithium ion transport in the polymer matrix and to enhance interfacial contacts between electrolyte and electrodes [7,8]. Nevertheless, development of PEO/lithium salt systems has been largely hampered by the following issues: (1) insufficient ionic conductivity to enable significant battery performance at a temperature below the melting point of 65 °C and relatively poor dimensional stability above the melting point; (2) low lithium ion transference numbers ($t_{Li^+} < 0.3$), giving rise to formation of a strong concentration gradient during battery operation

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with deleterious effects of dendritic growth and limited power delivery [9].

SIPE-based Li-ion batteries have been demonstrated recently to be thermally stable, mechanically robust and electrochemically conductive [10,11]. Unfortunately, virtually all the batteries rely on solvents contained in polymer matrices and pores in electrolyte membranes to carry out ion conduction between electrodes. It has been shown that a use of an appropriate solvent may boost the ionic conductivity of a membrane by 2–3 orders of magnitude in a battery cell via solvating ionic species, accelerating ion diffusion and enhancing interfacial contacts between electrodes and electrolyte. Unfortunately, to date, only a handful of the reported gel SIPE-based Li-ion batteries displays significant device activities at room temperature, although the SIPE materials are often of high ionic conductivities with high ion transference numbers [12]. The inactive battery performance at room temperature is largely attributed to the high interfacial resistance between electrodes and electrolyte membranes. The issue associated with the poor interfacial contacts becomes even more severe if no solvent is allowed in a membrane, which forces lithium ions to undergo not only hopping in the polymer matrix but also jumping from the electrolyte to electrodes, instead of diffusing between electrodes carried out by solvent molecules. This makes fabrication of all-solid-state Li-ion batteries technically highly challenging. Clearly, to avoid using a solvent in design of an all-solid-state battery, two important factors must be taken into account: (1) abundant anionic and electron rich species in an electrolyte membrane to facilitate Li-ion hopping in the polymer matrix; (2) a “soft” electrolyte membrane to promote Li-ion transport at electrolyte-electrode interfaces.

In a recent study, Bouchet and co-workers reported an elegant design and fabrication of the first all-solid-state SIPE based Li-ion battery with significant performance at above 60 °C. The electrolyte membranes were made of a triblock copolymer with a backbone containing various percentages of linear poly(ethylene oxide) (PEO) segments and side chains comprised of bis(trifluoromethanesulphonyl)imide lithium. PEO is flexible and becomes increasingly soft at elevated temperatures with a melting point of 65 °C [6,12]. The electron-rich oxygen atoms facilitate Li-ion hopping inside the polymer matrix and the softness of the membrane is beneficial for the ion transport at the electrolyte-electrode interfaces, resulting in remarkable device performance at 60 °C.

Here we report fabrication of an all-solid-state lithium-ion battery device that displays outstanding performance at an unprecedentedly low temperature of 50 °C using a non-fluorinated single-ion conducting polymer electrolyte (SIPE). This is an important step toward ambient temperature operation of SIPE-based battery devices. The SIPE material based on sp^3 boron central atom was synthesized via a reaction between a sp^3 boron based precursor (lithium tetramethanolatoborate) and stoichiometric bisphenol A. The SIPE membranes are then fabricated via eutectic mixing with PEO. An all-solid-state Li-ion battery with the membrane as the electrolyte was fabricated. Remarkably, the battery exhibits an excellent performance at a temperature as low as 50 °C, which is the lowest working temperature of a SIPE-based battery ever reported. Furthermore, the device displays highly stable operability at 80 °C, which overcomes the poor dimensional stability of PEO at high temperature above its melting point.

2. Experimental

2.1. Materials

Bisphenol A ($\geq 99\%$), dichloroethane, trimethyl borate, and

poly(vinylidene fluoride-hexafluoropropylene) (PVdF-HFP), were purchased from Sigma–Aldrich. 1,2-dichloroethane and acetonitrile were dried over phosphorus(V) oxide, distilled and stored over dried 4 Å molecular sieves. Methanol was dried over magnesium/iodine, distilled and stored over dried 4 Å molecular sieves. 3 wt.% of a drying agent and a solvent were added into a flask and then connected to the solvent processor. The top of the solvent processor was connected to a condenser, which was then linked to an argon inlet tube. Subsequently, the solvent was distilled for at least 6 h under argon atmosphere. After that, the pure solvent was collected using the solvent processor, followed by cooling and moving to a bottle. The material was then stored with a dried 4A molecular sieve. Bisphenol A was dried at 80 °C under reduced pressure overnight before use.

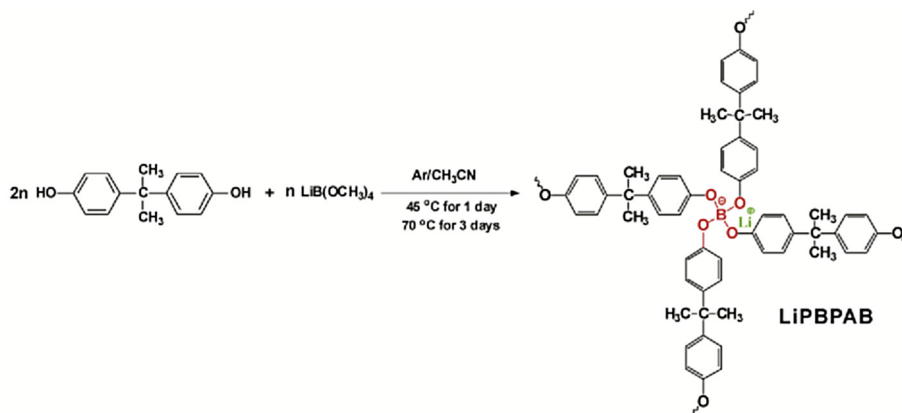
2.2. Synthesis and characterization of LiBPBAP

2.2.1. Synthesis of LiBPBAP

The LiBPBAP compound was synthesized and purified following the similar procedure reported previously [13,14]. A one-step process was employed to synthesize the designed sp^3 boron based SIPE (Scheme 1) by reacting $(LiB(OCH_3)_4)$ with stoichiometric bisphenol A (BPA) under argon atmosphere. Lithium tetramethanolatoborate was synthesized and purified following the procedure described in our previous studies [13–15]. Secondly, the stoichiometric bisphenol A (BPA) and the as-prepared lithium tetramethanolatoborate $(LiB(OCH_3)_4)$ precursors were added to a Pyrex flask containing dry acetonitrile (Baker's HPLC grade, dried) in an argon-filled glove box. The flask was then transferred to a fume hood and immediately connected to an argon flow line. The reaction was then carried out at 45 °C for 1 day and subsequently at 70 °C for another 3 days. The raw product was filtrated and washed continuously with acetonitrile. Finally, the product was dried at 120 °C under vacuum for 24 h for further characterization.

2.2.2. Characterization of LiBPBAP

To understand the electrostatic interaction between lithium ions and sp^3 boron atoms in LiBPBAP, density functional theory calculations were performed under the generalized gradient approximation (GGA) with the Pedrew-Wang exchange-correlation functional (PW91) using a double-numerical atomic basis set augmented with polarization functions. Molecular weight and polydispersity index (PDI) were measured by Gel Permeation Chromatography (GPC) (Waters 515 HPLC Pump, Waters 2707 Autosampler, Waters 2414 Refractive Index Detector). Pure water was used as an eluent at a flow rate of 0.8 ml min^{−1}. The poly(ethylene oxide) (PEO) standard was used for the calibration. All infrared spectra were taken with a Bio-Rad Excalibur FTIR spectrometer in the 400–4000 cm^{−1} frequency range. The ¹H and ¹¹B NMR spectra were recorded on a Bruker AMX (500) spectrometer at 500 MHz with trifloroborane (BF₃) in an ether solution (as the standard). Dimethyl sulfoxide-*d*₆ and deuterium oxide were used as solvents for the NMR test. The powder X-ray diffraction (XRD) experiment was performed on a D5005 Bruker AXS diffractometer with the sample size of 70–110 mg and Cu-K α radiation ($\lambda = 1.5410$) as the source at 40 kV voltage in the scanning range between 1.4° and 60° at room temperature. The morphologies of the polymer electrolytes were probed using the Scanning Electron Microscopy (SEM) with QUANTA 200 FEG. Samples were prepared via platinum sputtering under 5×10^{-2} mbar at room temperature (20 s, 30 mA) with a Baltec SCD050 apparatus. Tensile tests of the membranes were performed using the Instron universal materials testing system (model 5544) with a 10 N load cell in a constant relative humidity (50%) at 25 °C. Rectangular-shaped samples were cut from the membranes (5 mm wide with a gauge length of 10 mm). The



Scheme 1. The synthetic route of the lithium poly (bisphenol A borate) (LiBPAB).

thickness of the samples was measured with a digital micrometer having a precision of 1 μm . A cross-head speed of 10 mm/min was used. The enthalpy of melting for the SIPE membranes was determined using TA Instrument 2920. The experiment was performed under nitrogen atmosphere at the heating rate of 10 $^{\circ}\text{C}/\text{min}$. The thermal degradation study was performed under the inert atmosphere of N_2 (flow rate: 60 $\text{cm}^3 \text{min}^{-1}$) at the heating rate of 10 $^{\circ}\text{C} \text{min}^{-1}$ using the Thermogravimetric Analyzer (model TGA Q 50). Before the test, LiBPAB was dried overnight at 80 $^{\circ}\text{C}$ under vacuum. The thermal stability test was conducted from room temperature to 800 $^{\circ}\text{C}$. The melting enthalpy for SIPE membranes was determined using TA Instrument 2920. The analysis was performed under nitrogen atmosphere at the heating rate of 10 $^{\circ}\text{C}/\text{min}$.

2.3. Preparation of SIPE membranes

To fabricate a softer membrane and to enable battery operation at a lower temperature, we mixed PEO with the polymer by taking advantages of the flexibility and the electron-rich oxygen atoms of PEO that facilitate Li-ion conduction and reduce interfacial resistance. Nonporous and smooth sp^3 boron based SIPE membranes of five different contents of LiBPAB (denoted as S-BSM-1 - S-BSM-5)), which is necessary for preventing lithium dendrite growth and improving the power efficiency [12], were fabricated upon blending with a PEO binder by employing a temperature-dependent doctor blade method followed by fast evaporation of water at 120 $^{\circ}\text{C}$. The fast evaporation of water leads to a quick fabrication of membrane, which gives rise to the formation of highly smooth membranes with low porosity. The specific membrane casting procedure is as follows. A calculated amount of PEO and LiBPAB was dissolved in water at 80 $^{\circ}\text{C}$ to form an 8 wt. % solution of PEO and LiBPAB mixture. The composite polymer solution was subsequently cast onto a glass and bladed with a controlled thickness with the blade device. The glass was then transferred to a hot plate at 120 $^{\circ}\text{C}$ to vaporize the water. Since PEO is a hygroscopic material, it is difficult to remove the absorbed water from the pristine PEO, which may result in degradation of battery components and thus affect battery performance [16]. Therefore, in the present work, the prepared membranes were dried under vacuum at 80 $^{\circ}\text{C}$ for two days to ensure complete removal of water residues. Subsequently, a membrane was peeled off from the glass and stored in an argon-filled glove box (<1 ppm H_2O) for at least 1 week for further characterizations. The thickness of the prepared membranes ranges from 30 μm to 40 μm (Fig. 1). Finally, we note from Fig. 4 that a small amount of water might present in the membrane due to the exposure to air during the

sample preparation process prior to the test, which may positively affect the ionic conductivity. Upon drying, the water content is expected to be largely driven out and the battery was assembled in a glove box right after that. Thus, the effect of water on the battery performance is expected to be minimum.

2.4. Electrochemical characterization of SIPE membranes

The ionic conductivity of polymer electrolytes was measured by Electrochemical Impedance Spectroscopy (EIS) using the Zahner potentiostat-galvanostat electrochemical workstation model, PGSTAT, with the EIS module over a frequency range of 4×10^6 to 1 Hz and an oscillating voltage of 5 mV. ENREF_20A stainless steel cylindrical device of 1.5 cm diameter was used for conductivity measurement. The SIPE membrane was sandwiched between two stainless steel electrodes. The measurement device was sealed in a glove box under argon and heated at 80 $^{\circ}\text{C}$ in an oven to ensure a maximum contact between the membrane and the stainless steel electrodes prior to the determination of ionic conductivity. The Nyquist plot of imaginary vs real impedance was obtained and fitted with the Simulated Impedance Measurement (SIM) software offered by the EIS instrument. The electrochemical stability test (cyclic voltammetry) was conducted in the same stainless steel device using the CHI instrument in the voltage range of 0–7 V at a scan rate of 2 mV s^{-1} . A circular thin sheet of lithium metal along with the electrolyte membrane was placed inside the cavity of the device and sealed in a glove box under argon atmosphere. The lithium-ion transference number, t_{Li^+} , was measured for the Li/SIPE membrane/Li battery cell and calculated [17,18]. The value of the ion transference numbers of the membranes at 25 $^{\circ}\text{C}$ and 60 $^{\circ}\text{C}$ was derived from the combination of the complex impedance and potentiostatic polarization methods proposed by Evans et al. [19] using the Eq. (1):

$$t_{\text{Li}^+} = \frac{I_s(\Delta V - I_0 R_0)}{I_0(\Delta V - I_s R_s)} \quad (1)$$

where ΔV is the potential applied across the cell, I_0 and I_s are the initial and steady-state currents and R_0 and R_s are the initial and steady-state resistances of the passivation layers on the Li electrode.

2.5. Battery test

To evaluate the application potential of the S-BSM materials as all solid polymer electrolytes in lithium batteries, we assembled a

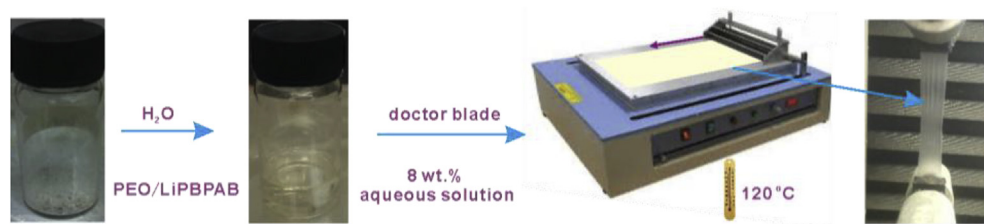


Fig. 1. The procedure for the preparation of the solid SIPE membranes.

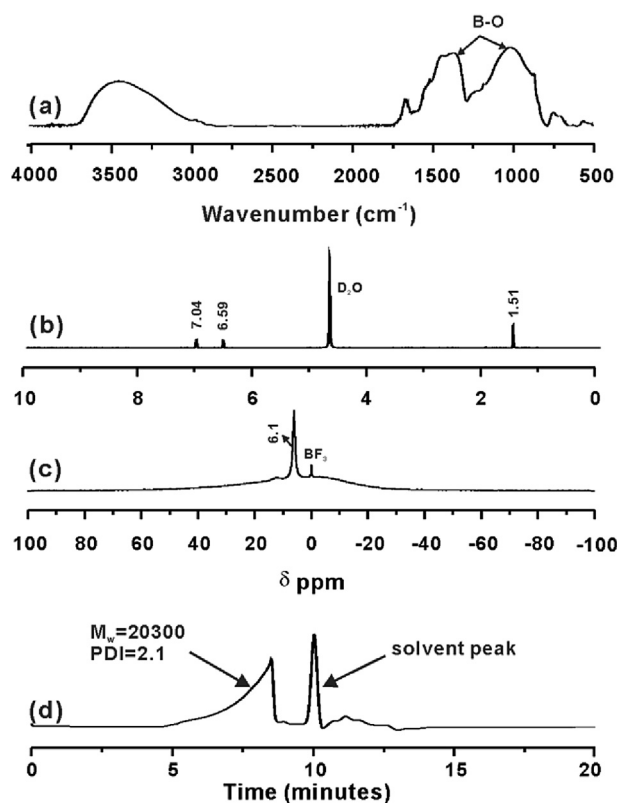


Fig. 2. (a) the FT-IR spectrum; (b) the ^1H NMR spectrum; (c) the ^{11}B NMR spectrum; (d) the GPC curve.

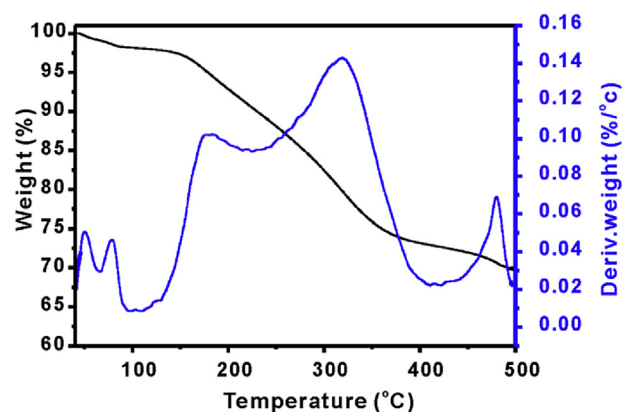


Fig. 4. The TGA graph and the derivative curve of the LiBPAB.

cathode capacity (C) in n hours. The composite cathode was prepared by casting a well stirred solution of LiFePO_4 (75 wt. %), PVdF (10 wt. %), acetylene black (10 wt. %) and a small amount of lithium bis(4-carboxy phenyl sulfonyl)imide [20,21] (5 wt. %) as the supporting electrolyte in a NMP solvent onto an aluminium foil. The resulting electrode was initially dried at 60 °C to vaporize the majority of the solvent and then kept at this temperature in a vacuum oven for 12 h to remove the solvent completely. The dried cathode was then cut into a circular shape used in coin cells. The assembling of the standard coin cells (CR2025) was done inside a glove box. A multichannel battery testing instrument Arbin BT-2000 was used to measure the discharge capacity of the coin cells assembled with the PEO/LiBPAB membrane.

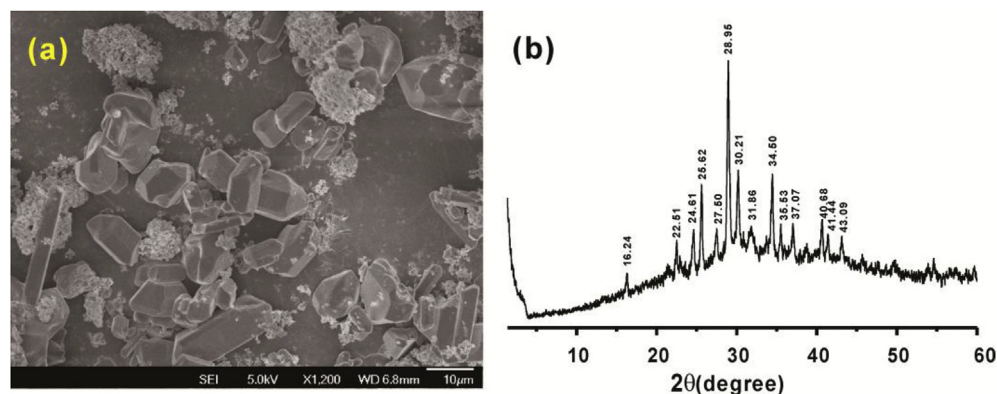


Fig. 3. (a) the SEM image and (b) the XRD spectrum of the LiBPAB.

$\text{Li}|\text{S-BSMs}|\text{LiFePO}_4$ cell and conducted cycling tests at various (C/n) rates, where C/n corresponds to a full discharge of the theoretical

3. Results and discussion

3.1. Characterizations

The FT-IR spectrum depicted that the characteristic absorption in the range of 1000 cm^{-1} to 1400 cm^{-1} confirms the formation of the B–O bonds (Fig. 2a) [22,23]. The ^1H NMR displays three peaks at 7.06, 6.57 and 1.63 ppm, arising from the BPA linkers (Fig. 2b). The ^{11}B NMR spectrum exhibits a sharp peak at 6.1 ppm, which falls in the range of 0–10 ppm, demonstrating that all boron atoms adopt a sp^3 configuration (Fig. 2c) [18,19,24]. The molecular weight of LiBPAB was obtained from the GPC analysis with DMF as the solvent. The number-average molecular weight (M_n) of the compound is 20,300 with the polydispersity index (M_w/M_n) of 2.1, suggesting a relatively narrow distribution of molecular weights (Fig. 2d) [18,19].

3.2. Morphology and microstructure characterizations

The SEM image (Fig. 3a) and the sharp diffraction pattern (Fig. 3b) of LiBPAB indicate a prismatic-like morphology with a length of approximately $10\text{ }\mu\text{m}$, suggesting that this material has a long range order [19].

3.3. Thermal stability of LiBPAB

Li-ion batteries usually display better electrochemical performance at elevated temperatures due to the enhanced overall kinetics. High electrochemical and thermal stabilities of electrolyte and electrode materials are thus required in the chosen temperature range. In general, conventional commercial Li-ion batteries are operated at temperature below $80\text{ }^\circ\text{C}$ because the polyolefin-based separators are thermally unstable [25,26] and a dual-ion electrolyte undergoes slow decomposition [27,28] at elevated temperatures [29,30]. The thermal stability of LiBPAB was determined by a thermogravimetric analysis method (TGA) and the TG curve is shown in Fig. 4. Clearly, the compound is thermally stable up to $145\text{ }^\circ\text{C}$, which surpasses the thermal stability requirement for electrolytes in Li-ion batteries. The excellent thermal stability of the electrolyte meets the requirement for high temperature operation.

3.4. Morphology, mechanical strength and electrochemical stability of the SIPE membranes of the SIPE membranes

The quality of a membrane is intimately dictated by the membrane structure. A rough porous membrane may give rise to unevenness of local current density at the electrolyte/anode interface during battery operation, which has been recognized as the main source of lithium dendrite growth [29,30]. The problem of dendrite formation can be largely overcome by employing a non-porous and uniform electrolyte membrane [31]. Fig. 5a and b show a highly uniform, smooth and non-porous morphology of the membrane surface and cross-section obtained with scanning electron microscopy (SEM).

The tensile stress/strain curves of the LiBPAB/PEO membranes made with the given components of active electrolyte materials are shown in Fig. 5c. The best tensile strength was found to be ca. 9.0 MPa for the $[\text{EO}]/[\text{Li}^+]$ ratio of 48, similar to what was reported previously on the triblock copolymer P(STFSiLi)-b-PEO-b-P(STDSiLi) membrane [12]. The excellent mechanical strength allows the membranes to readily sustain the redox reactions in battery charge and discharge processes with good longevity [32]. The tensile strength declines modestly as the LiBPAB component in the membrane increases. The strong mechanical strength is essential

for mitigating lithium dendritic growth.

The electrochemical stability of S-BSM-2 was analyzed by a cyclic voltammetry (CV) method $60\text{ }^\circ\text{C}$ in the range of 0–7 V (versus Li^+/Li) with a scanning rate of 2 mV/s (Fig. 5d). The membrane exhibits a stable electrochemical window of at least 5.5 V at both temperatures, which is significantly wider than that of the PEO/lithium bis(oxalate)borate (LiBOB) composite electrolyte ($\sim 4.5\text{ V}$) [33]. The extraordinary electrochemical stability is important for batteries with multi-component oxides, such as $\text{LiCu}_x\text{Mn}_y\text{O}_4$ [34] and $\text{LiCo}_x\text{Mn}_y\text{O}_4$ [35] as cathode materials, which possess significantly higher energy densities than conventional cathode materials, such as LiFePO_4 and LiMn_2O_4 , but have suffered from lack of appropriate electrolyte materials that match up with the required high voltage of the oxide compounds.

3.5. Thermal properties and ionic conductivities of the SIPE membranes

The thermal stability of the membranes was determined with differential scanning calorimeter (DSC) and the values were obtained from the measured data without normalization to the PEO content in the membrane. The membranes of the blended polymers with the given compositions, the equivalent molar ratios of $[\text{EO}]/[\text{Li}^+]$, the measured melting enthalpies and the calculated crystallinities are summarized in Table 1 and the DSC curves are shown in Fig. 6a. The endothermic peak of the neat PEO membrane at $61\text{ }^\circ\text{C}$, corresponding to the crystallinity melting, is modestly lower than the reported PEO melting point. Upon eutectic mixing with LiBPAB, the melting points of the blended membranes decline successively as the PEO component decreases. The reduction of melting points is largely attributed to the eutectic effect, which allows Li^+ ions to move from the polyanions to the PEO chains fluidly driven by the electrostatic interaction between Li^+ ions and the electron-rich oxygen atoms of the $-\text{CH}_2\text{CH}_2\text{O}-$ segments of the PEO matrix above the melting points. Indeed, the melting enthalpy displays a sharp decrease as the content of LiBPAB increases (Table 1), arising from the lower crystallinity caused by the higher concentration of the solvated Li^+ in the PEO matrix [12], similar to what was observed in other PEO/SIPE systems²⁰ and in PEO/solid oxide electrolytes [36]. We note that the PEO melting point can be further reduced by incorporating fillers such as two-dimensional graphene oxide nanosheets [37].

Fig. 6b depicts the temperature dependence of the ionic conductivity of the fabricated membranes obtained during the cooling process from $90\text{ }^\circ\text{C}$ to room temperature. As expected, the ionic conductivity increases as the $[\text{EO}]/[\text{Li}^+]$ ratio decreases. In particular, all ionic conductivity curves display an abrupt drop at around $60\text{ }^\circ\text{C}$, reflecting the well-known phase transition of PEO from the crystalline state to the amorphous state, accompanied by a sharp increase in ionic conductivity measured individually at a given temperature. The highest ionic conductivity was found to be $1.7 \times 10^{-7}\text{ S/cm}$ and $1.7 \times 10^{-4}\text{ S/cm}$ at $25\text{ }^\circ\text{C}$ and $60\text{ }^\circ\text{C}$, respectively, with the $[\text{EO}]/[\text{Li}^+]$ ratio of 8, about one order of magnitude higher than the highest value of a solid SIPE (around $1.2 \times 10^{-8}\text{ S/cm}$ at $25\text{ }^\circ\text{C}$ and $1.3 \times 10^{-5}\text{ S/cm}$ at $60\text{ }^\circ\text{C}$) ever reported [12].

3.6. Lithium ion transference number of the SIPE membranes

The steady-state current method was used to measure the Li ion transference number in a Li|blend polymer membranes|Li symmetric cell. The selected impedance spectra and the time-dependence response of DC polarization for the S-BSM-2 at $25\text{ }^\circ\text{C}$ are depicted in Fig. 6c and Fig. 7d. The measured values of the parameters used in Eq. (1) and the derived Li^+ transference numbers at both $25\text{ }^\circ\text{C}$ and $60\text{ }^\circ\text{C}$ are summarized in Table 2. The

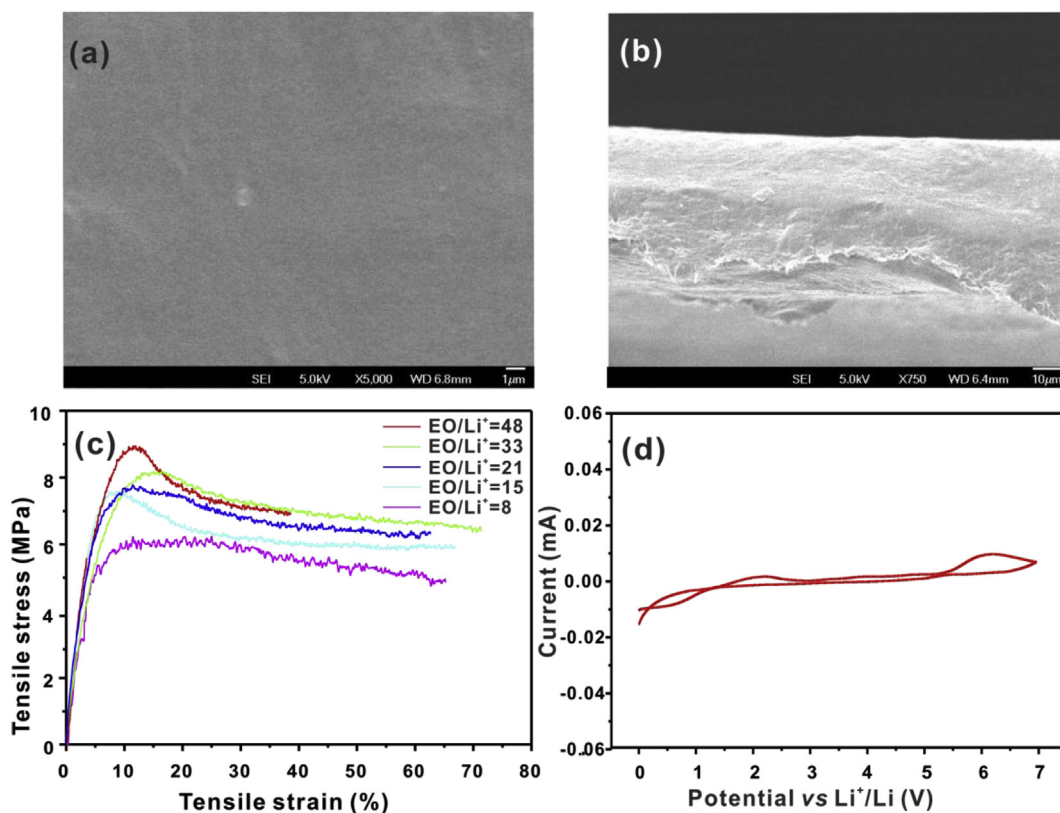


Fig. 5. (a) the SEM image of the S-BSM-2 surface, (b) the cross-section of S-BSM-2, (c) the mechanical strengths of S-BSMs, and (d) the electrochemical stability of S-BSMs at both room temperature and 60 °C.

Table 1

Compositions, equivalent molar ratios [EO]/[Li⁺], and the measured T_m , ΔH_f (PEO) and crystallinity of the prepared S-BSMs. The melting enthalpy values are normalized to the content of PEO.

Sample	Weight fraction of LiBPAB (%)	[EO]/[Li ⁺]	ΔH_f (PEO) (J g ⁻¹)	Crystallinity (%) ^a	T_m (°C)
S-BSM-1	57	8	31.2	18.6	52
S-BSM-2	40	15	45.9	27.4	55
S-BSM-3	33	21	57.0	34.0	58
S-BSM-4	25	33	91.6	54.6	59
S-BSM-5	18	48	100.4	65.9	60
neat PEO	—	—	167.6	100.0	61

^a The reference melting enthalpy of 100% crystalline PEO is 167.6 J g⁻¹, lower than the reported value [12].

calculated value of t_{Li^+} for S-BSM-2 is 0.92 at 25 °C and 0.96 at 60 °C. These values are higher than what were reported for the SIPE membranes by Bouchet et al. with bis(trifluoromethanesulfonyl) imide lithium salt (LiTFSI) attached to a flexible polyethylene backbone and close to unity [12]. The inorganic nano-particles such as nano-sized ceramic particles [38], metal organic frameworks (MOFs) [6,39], and core-shell silica particles [40] have been employed to increase lithium ion transference numbers of solid polymer electrolytes via the Lewis acid–base interactions between the surface of these fillers and the X⁻ anion of the salt. Unfortunately, the obtained transference numbers are usually much lower than the values of SIPEs which are close to unit. In particular, we note that the ion transference number decreases marginally at 60 °C, indicating that at the melting point the fluidity of the polymer increases. Nevertheless, the fluidity only gives rise to a slight increase of concentration gradient.

3.7. Battery test

The membranes of the S-BSM-2 electrolyte was assembled in coin cells with olivine (LiFePO₄) nanocrystals as the cathode and a Li foil as the anode for battery performance tests at different rates. Although room temperature operation of other PEO based all-solid-state lithium batteries with dual-ion electrolytes has been reported, it is the first time for an all-solid-state lithium battery with a single-ion PEO based electrolyte to operate with a discharge capacity of more than 100 mAh g⁻¹ at a temperature as low as 50 °C (Fig. 7a). Importantly, excellent cycling stability of more than 90 cycles is observed without significant capacity-fading regardless of the temperature (50, 60, 70 and 80 °C) and the given C-rate. In particular, the discharge capacity of 147 mA h/g at 80 °C is close to the theoretical value of 170 mA h/g for LiFePO₄ (Fig. 7a). The remarkable device performance is attributed to the excellent electrochemical and thermal stabilities of LiBPAB and the strong mechanical strength of the SIPE membrane. The results are comparable to the literature results with lithium-metal batteries using

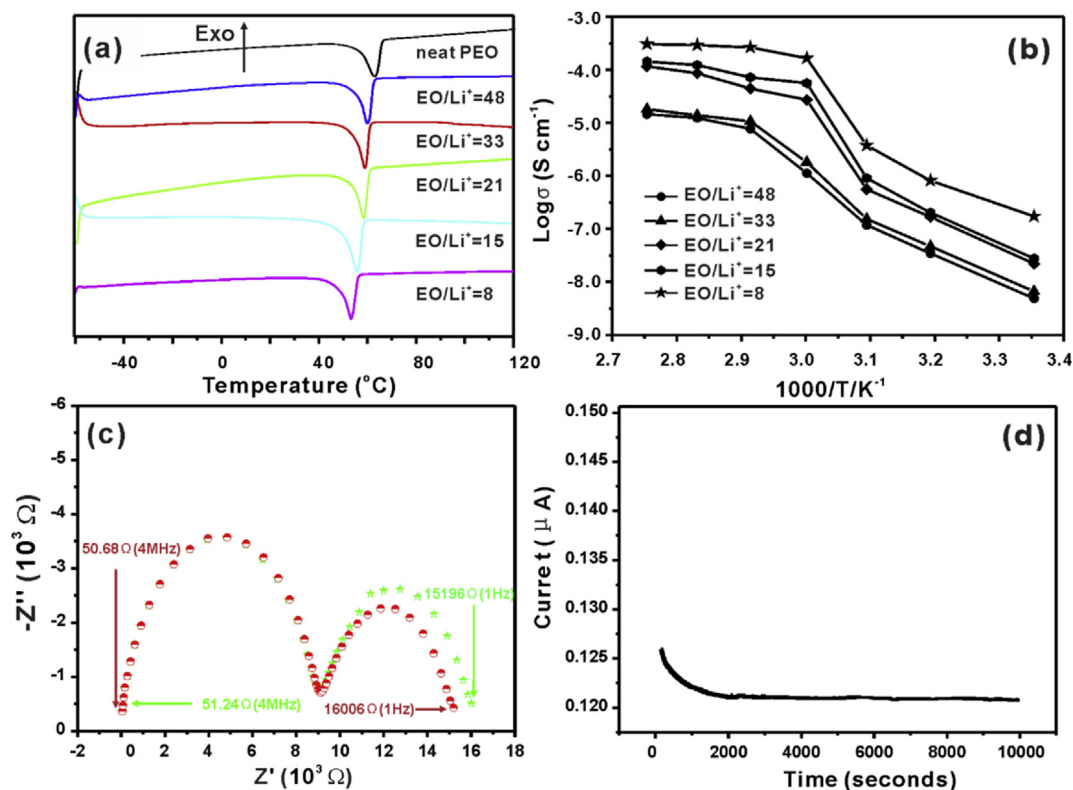


Fig. 6. (a) the DSC curves of S-BSMs, (b) the ionic conductivity of S-BSMs vs. the inverse of temperature, (c) impedance spectra and (d) time-dependence response of dc polarization for the S-BSM-2 at 25 °C.

solid SIPEs with BAB (P(STSILi)-PEO-P(STSILi)) triblock copolymeric structures [12] and solid polymer electrolytes (SPEs) with dispersed nano-composite S–ZrO₂ fillers [41]. Fig. 7b shows the characteristic evolution of the battery voltage at different discharge rates as a function of discharge capacity at 80 °C. The battery exhibits well-defined potential plateaus (>3.0 V) at all charge/discharge rates between C/25 and C/2.5 with the capacity retention of 147 mA h/g at C/25. The discharged capacity of these prototypes as a function of temperature is shown in Fig. 7c. The excellent power retention capacity at above 50 °C reveals remarkable performance compared to what has been observed with lithium-metal batteries using dry polymer electrolytes (Fig. 7c) [41]. The results at 50 °C are particularly interesting as they imply the possibility of a working temperature more than 10 °C below the melting point of PEO matrix. The interfacial resistances of the Li|S-BSMs|LiFePO₄ cell at several selected temperatures are depicted in Fig. 7d. A low interfacial resistance at 50 °C, similar to the resistance above the melting point of 60 °C for the PEO matrix, is readily visible, which might be attributed to the metastable state of S-BSMs between crystallization and recrystallization of the PEO matrix at this temperature. Indeed, in addition to the high ionic conductivity, the low interfacial resistance is also responsible for the remarkable battery performance at 50 °C. Furthermore, we note here that no battery activity was observed at room temperature due to the low ionic conductivity and the high interfacial resistance. All membranes experience only a slight decay of mechanical strength with a decrease of the [EO]/[Li⁺] ratio at room temperature (Fig. 7e), indicating that a low [EO]/[Li⁺] ratio is not necessarily at the expense of mechanical strength.

Recently, Claudio [6] reported an all-solid-state lithium battery assembled with a metal organic framework (MOF) laden PEO nanocomposite polymer electrolyte that displays significant

activity at 55 °C with improved ionic conductivity and lithium ion transference number. Very recently, Lee and coworkers assembled a hybrid solid electrolyte by blending PEO with tetragonal Li₇La₃Zr₂O₁₂ [36] and Li_{1.5}Al_{0.5}Ge_{1.5}(PO₄)₃ [42]. The all-solid-state lithium battery exhibits good capacity retention at temperatures below 55 °C. To overcome the drawbacks of PEO-based all-solid-state lithium batteries, several other polymer backbones were utilized to assemble all-solid-state batteries. For example, Lin [43] and coworkers prepared an electrolyte made of polysiloxane, poly(vinylidene fluoride) and lithium bis(trifluoro-methanesulfonyl)imide. Poly(vinylidene fluoride) offers outstanding mechanical strength and thermal stability and polysiloxane provides high ionic conductivity at ambient temperatures. Very recently, Mindemark [44] reported an electrolyte with a carbonate incorporated poly(ε-caprolactone) backbone. The material displays enhanced ionic conductivity and lithium ion transference number. These batteries exhibit superior room temperature cycle performance and appear to be considerably better than what was observed with the PEO based all-solid-state lithium batteries.

3.8. Mechanism of ionic conductivity

The high ionic conductivity and low interfacial resistance between the electrolyte membrane and the electrodes enable the battery assembled with a single ion electrolyte membrane to work in all solid state at an unprecedentedly low temperature, although the operating temperature is still well above the room temperature. In the LiBPAB compound, each of the sp³ boron atoms is covalently bonded with 4 oxygen atoms linked with electron-withdrawing aromatic linkers to facilitate charge delocalization, enabling Li-ions to be weakly associated with the polymer backbones [15]. This structure naturally leads to a weak electrostatic interaction

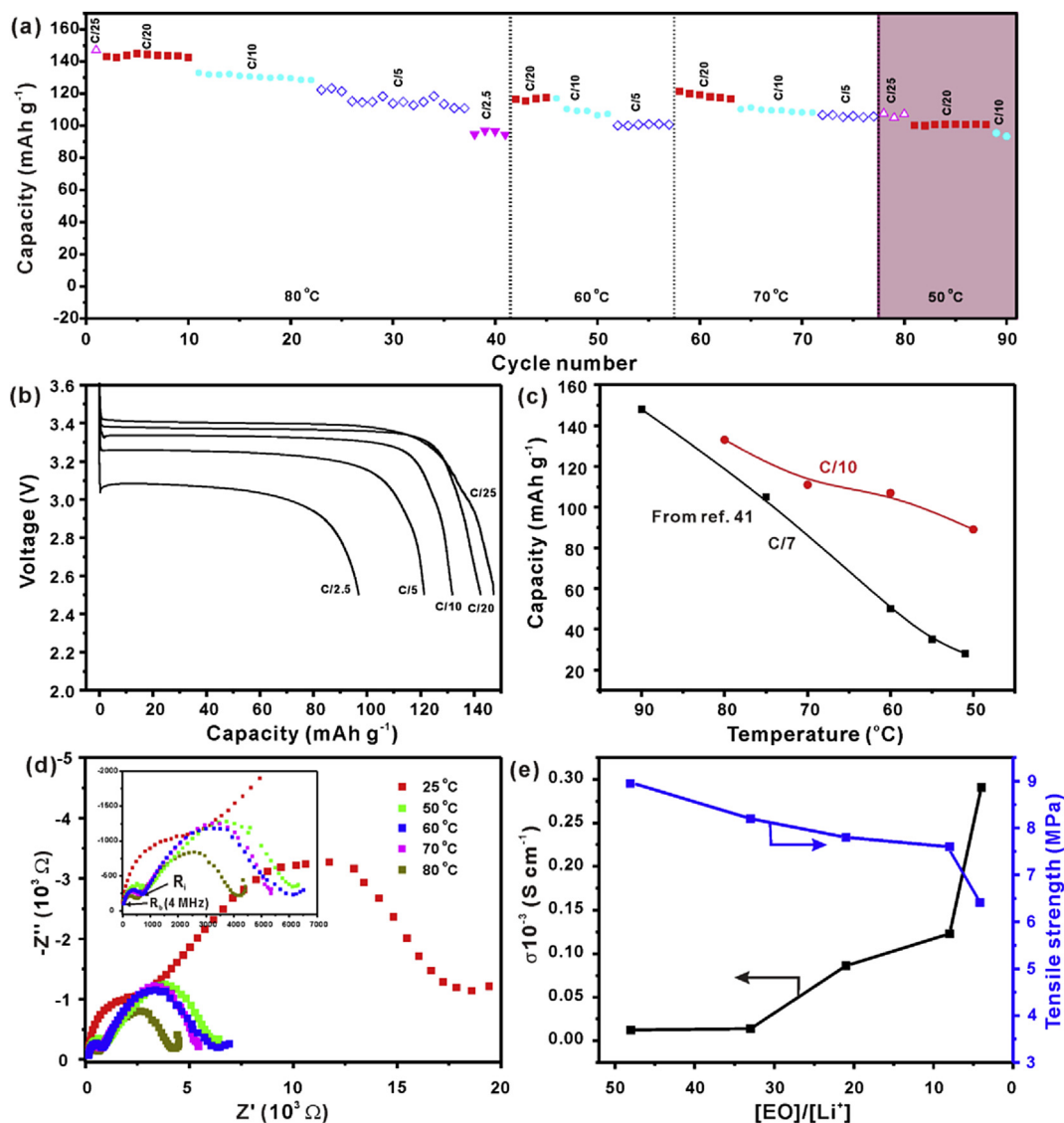


Fig. 7. (a) The cycle performance of the Li|S-BSM-2|LiFePO₄ battery in the temperature range of 50 °C–80 °C at various C rates; (b) the discharge profiles obtained at 80 °C at selected C-rates; (c) the temperature-dependence of discharge capacity. The data in red were taken from Fig. 3a and the data in black were taken from ref. [41]; (d) the Nyquist plots of a typical Li|S-BSM-2|LiFePO₄ at various temperatures; (e) the relationship between the ionic conductivity and the mechanical property of S-BSMs. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.).

Table 2

Measured values for the corresponding calculated values of lithium ion transference numbers (t_{Li}^+).

Temperatures	I_0 (μA)	I_S (μA)	R_0 (Ω)	R_S (Ω)	$\frac{I_1(\Delta V - I_0 R_0)}{I_0(\Delta V - I_S R_0)}$
25 °C	0.127	0.122	50.68	51.24	0.963
60 °C	2.45	2.26	5.84	5.91	0.922

between lithium ions and the anions covalently bonded to the polymers, which gives rise to high lithium ion mobility and high conductivity. Thus, the high ionic conductivity arises from the sp^3 configuration of boron and the bonding arrangements in the proximity of the boron atoms in the polymer, which allow the anionic charges on the boron atoms to be well dispersed in the polymer via the electron-withdrawing functionality of the aromatic linkers. As a consequence, the cationic lithium ions in the interstitials become weakly associated with the polymer, which thus intrinsically gives rise to high ion mobility [13,14]. Fig. 8a depicts

the electrostatic potential of LiBPAB calculated with density functional theory (DFT) under generalized gradient approximation. It reveals that the anionic charge residing on the boron atom is indeed delocalized in its vicinity in the polymer. Unfortunately, the anionic charge delocalization alone would not be able to substantially boost the ionic conductivity, as has been observed in several previous studies [13,14]. Several additional attributes of the polymer membranes are also required, which include.

1. polymers or blended polymers with sufficient flexibility to accommodate the motion of lithium ions in the matrix according to the free volume law [45–47].
2. a relatively low melting point of polymers or blended polymers, which raises the fluidity of the material to facilitates ion transport and to reduce electrolyte-electrode interfacial resistance;
3. electron rich segments in the polymers or the blended polymers to bridge lithium ions to promote ion hooping.

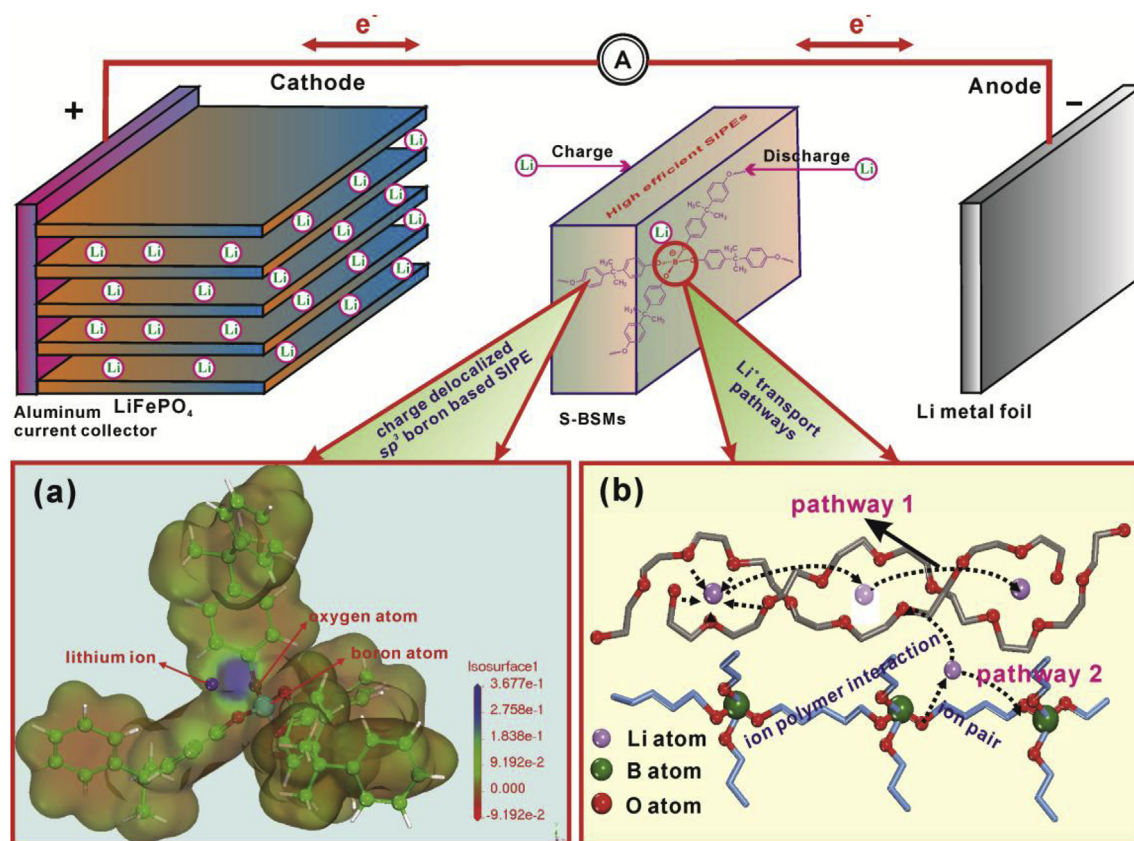


Fig. 8. (a) The calculated electrostatic potential of LiBPAB; (b) the illustrated lithium-ion transport pathways in S-BSMs.

The membrane formed by the single ion polymeric conductor, LiBPAB, blended with the PEO binder displays the required attributes. The basic structure of PEO chains separated Li^+ cations from the PBAB^- counteranions by coiled around them [48], therefore, PEO polymer requires local relaxation and segmental motion to allow Li^+ transport [3]. In the blended polymer, PEO/LiBPAB, the delocalization of the anionic charges arising from the sp^3 boron atoms anchored in the LiBPAB backbone gives rise to a high ion transference number of the membrane, while the PEO binder provides smooth passages for lithium ions in the blended polymer matrix. Fig. 8b illustrates two possible lithium ion pathways. One is hopping between the EO segments after leaving a sp^3 boron and another is EO-assisted hopping between sp^3 boron atoms. The interplay between LiBPAB and PEO leads to significantly lower melting points and melting enthalpies of the S-BSMs, substantially higher ionic conductivity and considerably better performance of an all-solid-state battery at a temperature as low as 50 °C.

4. Conclusion

In summary, we report an all-solid-state lithium-ion battery devised with an electrolyte membrane made of a sp^3 boron based polymer blended with PEO. The battery displays excellent performance at a temperature as low as 50 °C. By making use of the eutectic effect upon blending LiBPAB with PEO, sharply lower melting points and melting enthalpies of the electrolyte membranes, necessary for battery operation at near ambient temperatures, were achieved. A temperature-dependent doctor blade method was developed to fabricate smooth electrolyte membranes with excellent mechanical strength, strong thermal and electrochemical stability and high ionic conductivity. The interplay

between the polymeric electrolyte and PEO enables lithium ions to hop through the ethylene oxide (EO) segments or between the sp^3 boron atoms assisted by the EO segments. The measured ion transference numbers close to unity suggest that the electrolyte membranes exhibit virtually no concentration polarization in the electrochemical processes. The high lithium ion mobility arises from the weak association between lithium ions and the sp^3 boron atoms enabled by the anionic charge delocalization. Lithium ion transport in the polymer matrix is further facilitated by the EO segments of PEO. The interplay between LiBPAB and PEO enables lithium ions to hop through the EO segments or between sp^3 boron atoms assisted by the EO segments. This is an important step for fabrication of all-solid-state lithium-ion batteries toward operation at near ambient temperatures.

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References

- [1] J.M. Tarascon, M. Armand, *Nature* 414 (2001) 359–367.
- [2] S. Chandrashekar, N.M. Trease, H.J. Chang, L.-S. Du, C.P. Grey, A. Jerschow, *Nat. Mater.* 11 (2012) 311–315.
- [3] F. Croce, G.B. Appetecchi, L. Persi, B. Scrosati, *Nature* 394 (1998) 456–458.
- [4] M. Armand, J.M. Tarascon, *Nature* 451 (2008) 652–657.
- [5] J. Li, C. Ma, M. Chi, C. Liang, N.J. Dudney, *Adv. Energy Mater.* 5 (2015).
- [6] C. Gerbaldi, J.R. Nair, M.A. Kulandainathan, R.S. Kumar, C. Ferrara, P. Mustarelli,

- A.M. Stephan, J. Mater. Chem. A 2 (2014) 9948–9954.
- [7] M. Armand, Adv. Mater. 2 (1990) 278–286.
- [8] V. Di Noto, S. Lavina, G.A. Giffin, E. Negro, B. Scrosati, Electrochim. Acta 57 (2011) 4–13.
- [9] D.R. MacFarlane, J. Huang, M. Forsyth, Nature 402 (1999) 792–794.
- [10] Y. Zhu, S. Xiao, Y. Shi, Y. Yang, Y. Wu, J. Mater. Chem. A 1 (2013) 7790–7797.
- [11] X. Wang, Z. Liu, C. Zhang, Q. Kong, J. Yao, P. Han, W. Jiang, H. Xu, G. Cui, Electrochim. Acta 92 (2013) 132–138.
- [12] R. Bouchet, S. Maria, R. Meziane, A. Aboulaich, L. Lienafa, J.-P. Bonnet, T.N.T. Phan, D. Bertin, D. Gimes, D. Devaux, R. Denoyel, M. Armand, Nat. Mater. 12 (2013) 452–457.
- [13] Y. Zhang, Y. Sun, G. Xu, W. Cai, R. Rohan, A. Lin, H. Cheng, Energy Technol. 7 (2014) 643–650.
- [14] Y. Zhang, G. Xu, Y. Sun, B. Han, T.B. W. T. Z. Chen, R. Rohan, H. Cheng, RSC Adv. 3 (2013) 14934–14937.
- [15] Y. Zhang, R. Rohan, Y. Sun, W. Cai, G. Xu, A. Lin, H. Cheng, RSC Adv. 4 (2014) 21163–21170.
- [16] C. Xu, B. Sun, T. Gustafsson, K. Edstrom, D. Brandell, M. Hahlin, J. Mater. Chem. A 2 (2014) 7256–7264.
- [17] S.W. Feng, D.Y. Shi, F. Liu, L.P. Zheng, J. Nie, W.F. Feng, X.J. Huang, M. Armand, Z.B. Zhou, Electrochim. Acta 93 (2013) 254–263.
- [18] A. Ghosh, C.S. Wang, P. Kofinas, J. Electrochem. Soc. 157 (2010) A846–A849.
- [19] J. Evans, C.A. Vincent, P.G. Bruce, Polymer 28 (1987) 2324–2328.
- [20] Y. Zhang, R. Rohan, W. Cai, G. Xu, Y. Sun, A. Lin, H. Cheng, ACS Appl. Mater. Interfaces 6 (2014) 17534–17542.
- [21] Y. Zhang, J. Ting, R. Rohan, W. Cai, J. Li, G. Xu, Z. Chen, A. Lin, H. Cheng, J. Mater. Sci. 49 (2014) 3442–3450.
- [22] L. Larush-Asraf, M. Biton, H. Teller, E. Zinigrad, D. Aurbach, J. Power Sources 174 (2007) 400–407.
- [23] M.A. Ouis, A.M. Abdelghany, H.A. ElBatal, Process. Appl. Ceram. 6 (2012) 141–149.
- [24] I.M. Malkowsky, R. Frohlich, U. Griesbach, H. Putter, S.R. Waldvogel, Eur. J. Inorg. Chem. (2006) 1690–1697.
- [25] D. Djian, F. Alloin, S. Martinet, H. Lignier, J.Y. Sanchez, J. Power Sources 172 (2007) 416–421.
- [26] T.-H. Cho, M. Tanaka, H. Onishi, Y. Kondo, T. Nakamura, H. Yamazaki, S. Tanase, T. Sakai, J. Power Sources 181 (2008) 155–160.
- [27] A.A. Smagin, V.A. Matyukha, V.P. Korobtsev, J. Power Sources 68 (1997) 326–327.
- [28] S.E. Sloop, J.K. Pugh, S. Wang, J.B. Kerr, K. Kinoshita, Electrochim. Solid. St. 4 (2001) A42–A44.
- [29] T. Osaka, T. Homma, T. Momma, H. Yarimizu, J. Electroanal. Chem. 421 (1997) 153–156.
- [30] C. Brissot, M. Rosso, J.N. Chazalviel, S. Lascaud, J. Power Sources 81–82 (1999) 925–929.
- [31] W. Yi, Z. Huaiyu, H. Jian, L. Yun, Z. Shushu, J. Power Sources 189 (2009) 616–619.
- [32] T. Evans, J.H. Lee, V. Bhat, S.H. Lee, J. Power Sources 292 (2015) 1–6.
- [33] D. Zhang, H. Yan, H. Zhang, Z. Zhu, Qilu, Solid State Ion. 199 (2012) 32–36.
- [34] H. Kawai, M. Nagata, H. Tukamoto, A.R. West, Electrochim. Solid. St. 1 (1998) 212–214.
- [35] N. Kumagai, J.-M. Kim, S. Tsuruta, Y. Kadoma, K. Ui, Electrochim. Acta 53 (2008) 5287–5293.
- [36] J.H. Choi, C.H. Lee, J.H. Yu, C.H. Doh, S.M. Lee, J. Power Sources 274 (2015) 458–463.
- [37] S. Cheng, D.M. Smith, C.Y. Li, Macromolecules 48 (2015) 4503–4510.
- [38] F. Croce, R. Curini, A. Martinelli, L. Persi, F. Ronci, B. Scrosati, R. Caminiti, J. Phys. Chem. B 103 (1999) 10632–10638.
- [39] R.S. Kumar, M. Raja, M.A. Kulandainathan, A.M. Stephan, RSC Adv. 4 (2014) 26171–26175.
- [40] J. Shim, D.-G. Kim, H.J. Kim, J.H. Lee, J.-C. Lee, ACS Appl. Mater. Interfaces 7 (2015) 7690–7701.
- [41] F. Croce, S. Sacchetti, B. Scrosati, J. Power Sources 162 (2006) 685–689.
- [42] Y.C. Jung, S.M. Lee, J.H. Choi, S.S. Jang, D.W. Kim, J. Electrochem. Soc. 162 (2015) A704–A710.
- [43] Y. Lin, J. Li, Y. Lai, C. Yuan, Y. Cheng, J. Liu, RSC Adv. 3 (2013) 10722–10730.
- [44] J. Mindemark, B. Sun, E. Törmä, D. Brandell, J. Power Sources 298 (2015) 166–170.
- [45] M.R. Johan, L.M. Ting, Int. J. Electrochem. Sci. 6 (2011) 4737–4748.
- [46] H. Nithya, S. Selvasekarapandian, P.C. Selvin, D. Arun Kumar, M. Hema, Ionics 17 (2011) 587–593.
- [47] J.L. Souquet, F. Alloin, O. Brylev, M. Duclot, J.Y. Sanchez, Ionics 4 (1998) 1–7.
- [48] P. Lightfoot, M.A. Mehta, P.G. Bruce, Science 262 (1993) 883–885.